Orchid Majumder

+1 425 241 1741 | orchidmajumder@gmail.com | LinkedIn | GitHub | Google Scholar | Seattle, USA

Work Experience

Google Workspace — Kirkland, USA

Senior Staff Software Engineer, Gemini Agent — *July 2026 – current*

- Working on long-horizon agentic evaluation and agentic recursive self-improvement for Gemini Agent on Google Workspace.

Amazon AGI — Seattle, USA

Principal Research Engineer, Amazon Nova (Multi-Modal Pre-Training) — *October 2023 – July 2026*

- Research engineering lead for Amazon Nova foundational models pre-training. Developed a *NeMo-Megatron* based distributed-training stack that is used to train Nova models *(using 5D parallelism)*.
- Worked on architecture research and pre-training scaling laws for Nova-2 and beyond.

AWS AI Labs — Seattle, USA

Senior Research Engineer, Diffusion Models / Computer Vision — *April 2019 – September 2023*

- Research engineering lead for Amazon Titan multi-modal models and text-to-image/video diffusion models. Developed a *PyTorch-FSDP* based distributed-training stack that was used to train large-scale image/video diffusion models on AWS.
- Research engineering lead for Amazon Rekognition Custom Labels, an AWS AutoML service aimed towards helping customers build their own ML models with a few labeled samples.
- Worked on research topics related to few-shot, continual and meta-learning algorithms for visual recognition and hyperparameter optimization *(see Google Scholar)*.

AWS SageMaker — Seattle, USA

Research Engineer, AI Labs — *March 2017 – March 2019*

- Part of the core team that launched Amazon SageMaker at AWS re:Invent 2017, a fully managed AWS service for building, training and deploying any kind of ML models.
- Led the development effort for Inference Pipeline for Amazon SageMaker.
- Developed deep-learning based NLP algorithms including Seq2Seq and Object2Vec which are offered as built-in algorithms within Amazon SageMaker.

Amazon — Bangalore, India

Software Engineer, Amazon India — *July 2014 – March 2017*

- Engineering lead for registration and order management for individual sellers on Amazon India — led a large-scale cross-team effort to develop a software system that enabled individual sellers to get started as professional sellers.
- Built a rule-based recommendation engine to help sellers determine the price for their products, repurposed Amazon logistics stack for individual sellers and simplified the registration workflow for new individual sellers.

Internship Experience

Amazon — Bangalore, India

Software Engineer Intern, Amazon Retail — *June 2013 – July 2013*

- Built an intermediate caching layer using Amazon DynamoDB to reduce request latency.
- Built a distributed log parser using Hadoop/MapReduce on Amazon EMR.

Indian Statistical Institute — Kolkata, India

Research Intern, Machine Perception / Computer Vision — *June 2012 – July 2012*

- Worked on fuzzy neural-network based algorithms to identify Indian dance styles from images.

Selected Publications

The Amazon Nova Family of Models: Technical Report and Model Card (arXiv) – Amazon AGI.

On the Scalability of Diffusion-Based Text-to-Image Generation (CVPR 2024) – Hao Li, Yang Zou, Ying Wang, Orchid Majumder, Yusheng Xie, R Manmatha, Ashwin Swaminathan, Zhuowen Tu, Stefano Ermon, Stefano Soatto.

Revisiting Contrastive Learning for Few-Shot Classification (arXiv) – Orchid Majumder, Avinash Ravichandran, Subhransu Maji, Marzia Polito, Rahul Bhotika, Stefano Soatto.

Incremental Meta-Learning via Indirect Discriminant Analysis (ECCV 2020) – Qing Liu, Orchid Majumder, Avinash Ravichandran, Rahul Bhotika, Stefano Soatto.

MARTHE: Scheduling the Learning Rate Via Online Hypergradients (IJCAI 2020) – Michele Donini, Luca Franceschi, Orchid Majumder, Massimiliano Pontil, Paolo Frasconi.

d-SNE: Domain Adaptation Using Stochastic Neighborhood Embedding (CVPR 2019 Oral) – Xiang Xu, Xiong Zhou, Ragav Venkatesan, Gurumurthy Swaminathan, Orchid Majumder.

Conferences

Area Chair – NeurIPS, ICML, ICLR, CVPR, ECCV.

Outstanding Reviewer – NeurIPS, CVPR, ECCV.

Peer Review – NeurIPS, ICLR, CVPR, ICML, ECCV, ICCV.

Journal Review – TPAMI, TMLR, IEEE TCSVT.

Patents

Automated Machine Learning Pipeline Exploration and Deployment (US : 11727314B2).

Building Sequence to Sequence (S2S) models using Previously Generated S2S Models (US : 11748610B1).

Academics

Columbia University in the City of New York New York, USA
M.S. in Computer Science (Machine Learning Track)

Jadavpur University Kolkata, India
B.S. in Computer Science and Engineering

Open Source

adatune | *GitHub*

- Implementation of IJCAI 2020 paper - "MARTHE: Scheduling the Learning Rate Via Online Hypergradients".

sagemaker-sparkml-serving-container | *GitHub*

- A toolkit for deploying SparkML models efficiently in SageMaker without an Apache Spark runtime.

Other Achievements

Mathematics Olympiad – One of the 500 students all over India to qualify for the Indian National Mathematical Olympiad (INMO) in 2008.

Entrance Examinations – Ranked in the top 0.1% in the secondary examination, higher-secondary examination and engineering entrance examination in the state of West Bengal in India.